BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLY PROGRESS REPORT (08/06/2026 – 14/06/2026)

WEEK 4

PROJECT NAME	Prithvi EO Models (IBM Model)

PROJECT DESCRIPTION :	**Fine tuning and Training of IBM Prithvi EO Models for Earth Observation tasks.**
GROUP MEMBER :	**Dhanya R, Ch Siddartha**
GROUP ID :	**2026G264**
GROUP GUIDE :	**Dr. Abdul Jhummerwala**
GROUP COORDINATOR :	**Dr. Abdul Jhummerwala**
COLLEGE GUIDE NAME :	**Dr. Shweta Saharan**
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COLLEGE NAME :	**Gati Shakti Vishwavidyalaya, Vadodara**

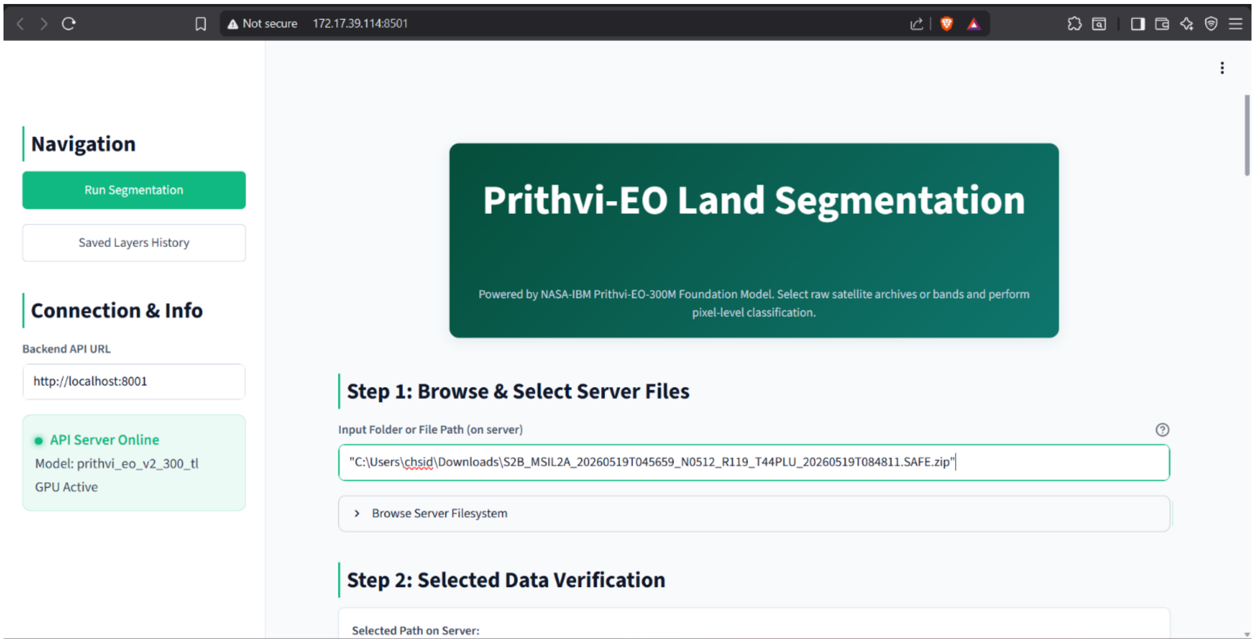
08/05/2026	• Refined the live execution pipeline status tracking. • displaying live checkbox status in the run segmentation pipeline.
09/05/2026	• changing and updating the UI for the Prithvi land classification.
10/05/2026	• Add a comparison module allowing users to load two historical layers (from different dates) from the database and compute pixel-wise change detection.
11/05/2026	• Develop a color picker UI in Streamlit that dynamically generates a Styled Layer Descriptor (SLD) XML file.
12/05/2026	• Making request to geoserver to view output layers instead of reading it from local cog file.
13/05/2026	**Holiday (2nd Saturday)**
14/05/2026	**Holiday (Sunday)**

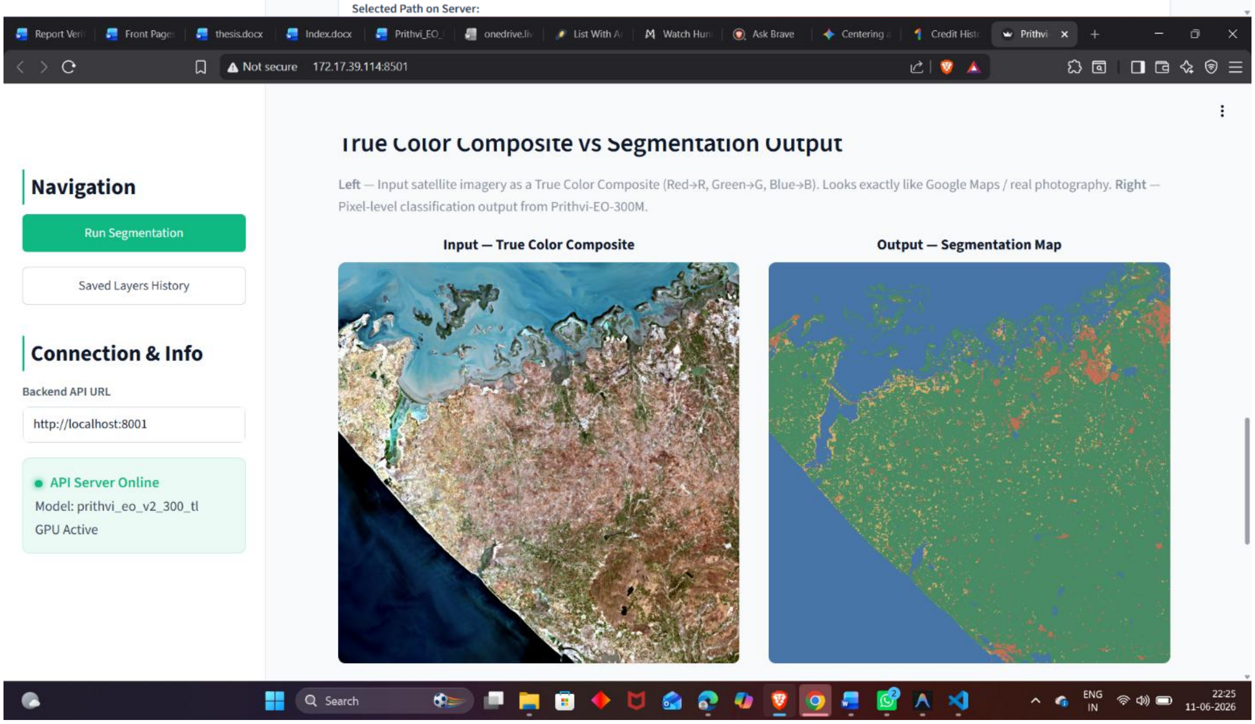
Next Week Plan	Preparing and Verificaion of Report, presenting our project.

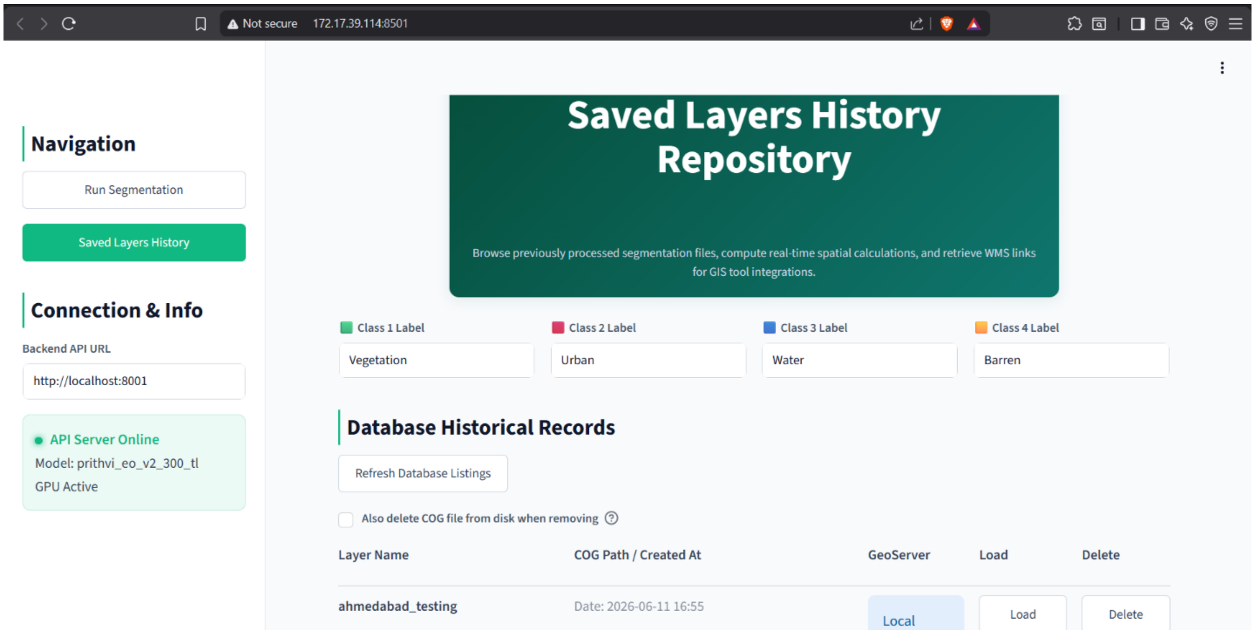
REFERENCE:

- ibm-nasa-geospatial (IBM-NASA Prithvi Models Family)
- https://www.youtube.com/live/CB3FKtmuPI8?si=oIB4EmCtlkUjSo0r
- Sentinel-2 | Copernicus Data Space Ecosystem
- https://research.ibm.com/blog/terramind-prithvi-tiny-small-models-geospatial
- https://github.com/NASA-IMPACT/Prithvi-EO-2.0

SCREEN SHOTS:






Not secure 172.17.39.114:8501
Navigation
Run Segmentation
Saved Layers History
Connection & Info
Backend API URL
http://localhost:8001
API Server Online
Model: prithvi_eo_v2_300_tl
GPU Active
Saved Layers History Repository
Browse previously processed segmentation files, compute real-time spatial calculations, and retrieve WMS links for GIS tool integrations.
Class 1 Label
Vegetation
Class 2 Label
Urban
Class 3 Label
Water
Class 4 Label
Barren
Database Historical Records
Refresh Database Listings
Also delete COG file from disk when removing
Layer Name
COG Path / Created At
GeoServer
Load
Delete
ahmedabad_testing
Date: 2026-06-11 16:55
Local
Load
Delete